Astrocytic ACSBG1 depletion improves lipid–cytokine signaling and attenuates α-synuclein pathology in a Parkinson's disease mouse model

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BACKGROUND

- Lewy pathology in Parkinson's disease (PD) contains phosphorylated α-synuclein; total and pS129-α-Syn levels track disease severity.
- Our genome-wide screens identified ACSBG1, an acyl-CoA synthetase, as an in vivo regulator of α-Syn. In brain it is expressed only in astrocytes.
- How astrocytic lipid metabolism is coupled to inflammatory activation and to neuronal α-Syn pathology is unknown.

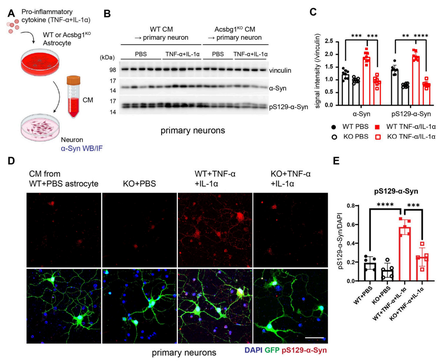
HYPOTHESIS

ACSBG1 enzymatic activity in reactive astrocytes shapes a secretome of cytokines and long-chain sphingosines that drives α-Syn accumulation and Ser129 phosphorylation in neurons.

APPROACH

- WT and Acsbg1 KO astrocytes + TNF-α/IL-1α → conditioned medium (CM) onto primary neurons
- Astrocyte RNA-seq, CM cytokine array, CM lipidomics
- Add-back, Triacsin C, enzyme-dead ACSBG1 (K701A)
- Acsbg1 KO × Thy1-α-Syn (Line 61) mice

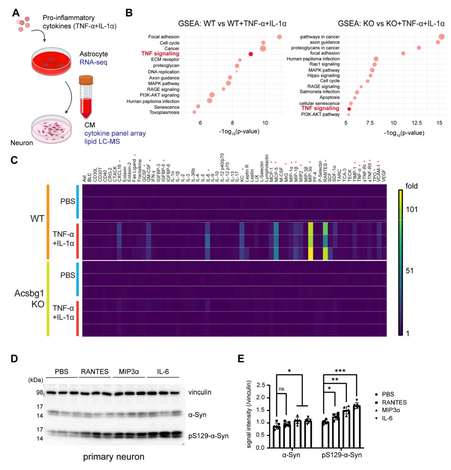
Fig. 1 KO medium fails to induce α-Syn pathology



(A) CM transfer design. (B, C) α-Syn and pS129-α-Syn immunoblot, normalised to vinculin; n = 8. (D, E) pS129-α-Syn (red) ICC, GFP (green), DAPI (blue); n = 5–6. Scale bar 50 µm. **p<0.01, ***p<0.001, ****p<0.0001.

Activated WT medium raised neuronal α-Syn and pS129-α-Syn. Activated KO medium did not.

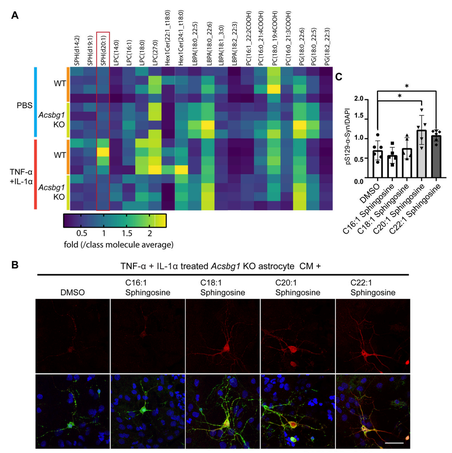
Fig. 2 Acsbg1 loss blunts the secretome



(A) Workflow: astrocyte RNA-seq plus CM cytokine array and lipid LC-MS. (B) GSEA, PBS vs TNF-α/IL-1α; TNF signalling (red) falls in rank in KO. (C) CM cytokine-array heatmap (1–101 fold). (D, E) Neurons given KO CM plus RANTES, MIP-3α or IL-6; n = 6. *p<0.05 to ***p<0.001; ns.

KO astrocytes release far less IL-6, RANTES and MIP-3α. Each cytokine alone is sufficient to raise neuronal pS129-α-Syn.

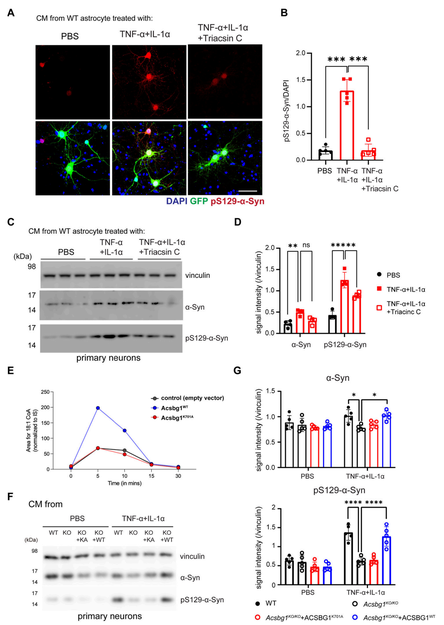
Fig. 3 ACSBG1 controls sphingosine secretion



(A) CM lipidomics heatmap, fold change vs class average; SPH(d20:1) boxed red. (B, C) KO CM supplemented with C16:1–C22:1 sphingosine; pS129-α-Syn / DAPI, n = 5–6. Scale bar 50 µm. *p<0.05.

SPH(d20:1) rises on activation and normalises in KO. Add-back of C20:1 / C22:1 — not C16:1 / C18:1 — reinstates neuronal pS129-α-Syn.

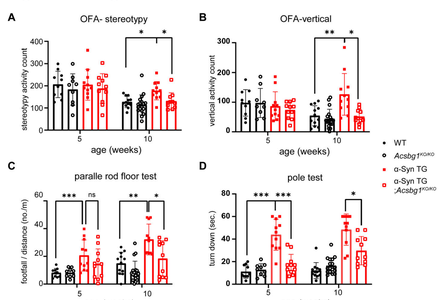
Fig. 4 Enzymatic activity is required



(A–D) Triacsin C (pan-ACS inhibitor) on astrocytes; ICC n = 5–6, immunoblot n = 3. (E) ACS activity of WT vs K701A ACSBG1 on C18:1. (F, G) KO astrocytes re-expressing WT or enzyme-dead ACSBG1; n = 5. *p<0.05 to ****p<0.0001; ns.

Triacsin C blocks the induction. WT ACSBG1 restores it in KO astrocytes; catalytically dead K701A does not.

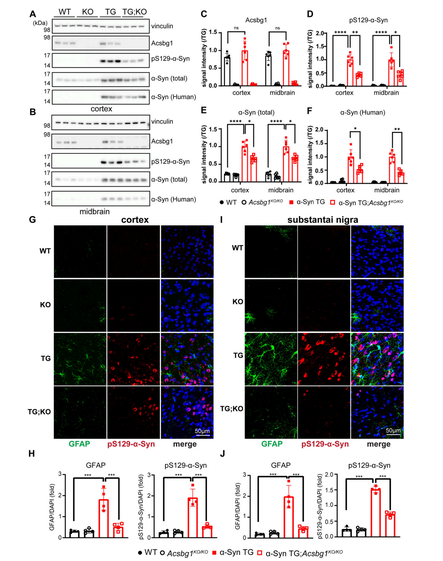
Fig. 5 Acsbg1 KO rescues motor behaviour



(A) Stereotypy and (B) vertical activity, open field. (C) Parallel rod floor test. (D) Pole test. 5 and 10 weeks; n = 9–18 per genotype. All male — the transgene is X-linked. *p<0.05, **p<0.01, ***p<0.001.

At 10 weeks TG;KO mice normalised stereotypy and vertical activity and outperformed TG mice on both motor tasks.

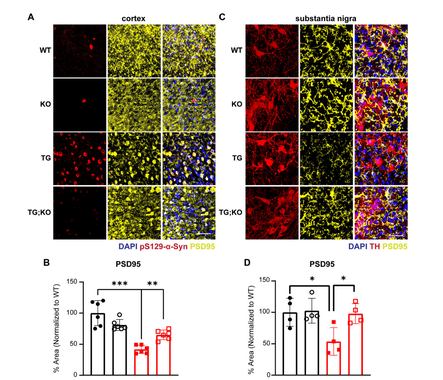
Fig. 6 Less α-Syn and astrogliosis in vivo



(A–F) Cortex (A) and midbrain (B) immunoblots at 20 weeks; ACSBG1, pS129-α-Syn, total and human α-Syn normalised to TG; n = 6. (G–J) GFAP / pS129-α-Syn staining in cortex and substantia nigra; GFAP / DAPI, n = 4. Scale bar 50 µm. *p<0.05 to ****p<0.0001.

Raised α-Syn and pS129-α-Syn in TG cortex and midbrain, and GFAP+ astrogliosis, are both attenuated on the Acsbg1-null background.

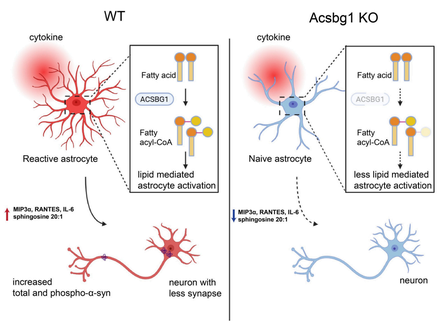
Fig. 7 Acsbg1 KO preserves synapses



(A, B) Cortex and (C, D) substantia nigra at 20 weeks; PSD95 with pS129-α-Syn or TH; PSD95 / DAPI, n = 4–6. Scale bar 50 µm. *p<0.05, **p<0.01, ***p<0.001.

PSD95 loss in TG cortex and substantia nigra is prevented by Acsbg1 deletion.

Model



ACSBG1 activity in reactive astrocytes licenses parallel release of cytokines and sphingosine d20:1, driving neuronal α-Syn accumulation, pS129 and synapse loss.

CONCLUSIONS

1. ACSBG1 acts non–cell-autonomously on neuronal α-Syn.
2. Two parallel mediators — IL-6 / RANTES / MIP-3α and sphingosine d20:1 — each sufficient on its own.
3. The effect requires ACSBG1 enzymatic activity.
4. Acsbg1 deletion in Thy1-α-Syn mice cuts astrogliosis and α-Syn, preserves PSD95 and rescues motor behaviour.
5. ACSBG1 expression is restricted to brain, adrenal gland, ovary and testis — an attractive PD target.

ONGOING WORK

- Necessity: cytokine-neutralising antibodies; blockade of astrocytic sphingolipid synthesis
- Aggregation: soluble / insoluble fractions, native gel, MJFR-14
- Production vs clearance; mitochondrial function and ROS

SELECTED REFERENCES

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